

Transactional Dynamics Between Parental Responsiveness and Child Emotion Dysregulation: A Longitudinal Study From Infancy to Early School Age

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Bidirectional influences between parenting and children's emotion regulation are well established, but developmental shifts in these dynamics and differences between mother– and father–child relationships are far from understood. We examined such bidirectional dynamics from infancy to early school age in 102 U.S. Midwestern community families (51 girls), using an autoregressive latent trajectory model that enabled us to distinguish within-dyad co-regulatory processes from traitlike stability across dyads. Parental responsiveness and child emotion (dys)regulation were coded from observed parent–child interactions at seven time points from 7 months to 6.5 years. Results demonstrated significant parent-to-child effects during toddlerhood in both mother– and father–child dyads, with higher parental responsiveness predicting better subsequent emotion regulation in children. However, child-to-parent effects were observed only in father–child dyads, such that children with poorer emotion regulation elicited more, and those with better emotion regulation elicited less paternal responsiveness at the later time point. These findings suggest fathers may adjust caregiving more flexibly, balancing recognition of children's emotional needs and of their growing autonomy, whereas maternal responsiveness may be less influenced by fluctuations in child emotion (dys)regulation. No significant bidirectional associations were observed in infancy or early school age. Findings suggest that bidirectional dynamics are developmentally fluid in early parent–child relationships and that, surprisingly, fathers may be more adept at calibrating their responsiveness based on children's regulatory needs.

Keywords: parental responsiveness, emotion dysregulation, autoregressive latent trajectory model, longitudinal study

Supplemental materials: <https://doi.org/10.1037/emo0001588.supp>

Dyadic social interactions are key contexts for the development of most aspects of human functioning, including children's regulatory capacities broadly and emotion regulation—key for adaptation and mental health—specifically (Morris et al., 2018; Sroufe, 1996). Emotion regulation consists of external and internal processes of monitoring, managing, controlling, and modifying emotional reactions in accordance with situational or individual demands (Thompson, 1994). For young children, parent–child interactions are fundamental settings in which to study the dyadic nature of emotion regulation, for several reasons. Family contexts are often affectively charged and infused with diverse emotions; parents can

influence children's emotion regulation through strategies such as validating or dismissing child emotions and modeling or coaching regulatory behaviors; and parent–child relationships are the basis of attachment relationships, critical for children's emotion regulation (Eisenberg et al., 1998; Morris et al., 2018; see the special issue of *Developmental Psychology*, Spinrad et al., 2020).

One key issue in understanding these dynamics—indeed, a central question in developmental psychology more broadly—is the direction of effects: Do they primarily flow from the parent to the child or from the child to the parent? Historically, in early research, children were often pictured as passive recipients of parental

This article was published Online First September 18, 2025.

Dylan Gee served as action editor.

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The analysis syntax can be found on the Open Science Framework (Kim, 2025; https://osf.io/j938r/?view_only=21cbbbc0f17d4254ab5568ac81863d4d), and the coding systems are available upon request. But the data supporting the current findings are not publicly available. The consents that participants signed precluded any sharing of individual data, even if deidentified.

The authors declare no conflicts of interest. This work was funded by grants from the National Institute of Mental Health, National Institutes of Health (Grants R01 MH063096 and K02 MH001446) and the Eunice Kennedy

Shriver National Institute of Child Health and Human Development, National Institutes of Health (Grant R01HD110427) awarded to Grazyna Kochanska. The authors thank all Child Lab team members for their contributions and all the participating families for their commitment to their research.

Juyoung Kim played a lead role in formal analysis, writing—original draft, and writing—review and editing and an equal role in conceptualization. Grazyna Kochanska played a lead role in funding acquisition and supervision, a supporting role in writing—review and editing, and an equal role in conceptualization and writing—original draft.

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influences. For the last few decades, however, typically traced to Bell's (1968) influential article, a perspective viewing children as active agents, which more recently evolved into one that highlights the bidirectionality of parent-child relations, has been largely advocated and accepted, galvanizing divergent, increasingly sophisticated methodological approaches (e.g., Cummings & Schermerhorn, 2003; Davidov et al., 2015; Kuczynski & De Mol, 2015; Loulis & Kuczynski, 1997; Martin, 1981; Scaramella & Leve, 2004; Wass et al., 2024).

In addition to the bidirectionality, researchers have long emphasized the need to integrate processes of developmental change into the study of socialization and parenting (Collins & Madsen, 2003; Maccoby, 1984b). The dynamics of interactions within parent-child relationships can be altered as children's needs and demands change with growing maturity and capabilities, and as parents adapt their goals, expectations, and behaviors in response to those changes. Research on children's developing emotion (dys)regulation is an especially compelling area in which to study changing dynamics of parent-child bidirectionality over time, with dynamics for regulatory abilities gradually shifting from parental regulation to co-regulation to child self-regulation (Collins & Madsen, 2003; Maccoby, 1984a; Sroufe, 1996).

Informed, in part, by attachment theory, we propose a model of adaptive bidirectional socialization dynamics. The model involves two aspects. One, the parent's responsive care promotes the child's future emotion regulation (parent-to-child effects); and two, adding nuance to those dynamics, the child's emotion regulation capacities determine, in part, the degree of the parent's future responsiveness, with some children needing more and some needing less parental support (child-to-parent effects).

Those adaptive dynamics are guided by the parent's and the child's implicit and explicit representations of each other, formed over the course of their relationship. Early parental consistent care promotes the child's emotion regulation, because the child comes to accurately expect comfort and support (Cassidy, 2021; Cassidy et al., 2025; Dykas & Cassidy, 2011; Main, 1990). The child's varying emotion regulation capacities determine the parent's appropriate level of response, because the parent can accurately gauge the child's psychological cues, needs, resources, and need for support (Garon-Bissonnette et al., 2025; Nieto-Retuerto et al., 2024; Trepiak et al., 2025).

Extensive empirical research has established the parent-to-child adaptive effects. Parental responsiveness has been implicated as critical for fostering adaptive emotion regulation in children, as responsive and sensitive parents recognize and react to children's emotional and behavioral cues promptly, accurately, warmly, and effectively (Ainsworth et al., 1978). Responsive, sensitive, warm, and supportive parenting contributes to children's better emotion regulation, whereas unsupportive, unresponsive, chaotic, and coercive parenting is related to poorer emotion regulation (see Beliveau et al., 2023; Goagoses et al., 2023; Paulussen-Hoogbeem et al., 2007; Wass et al., 2024 for reviews). These associations have been replicated across cultures (Haslam et al., 2020; Song et al., 2023), across measures, including physiology, neuroimaging, behavior, and reports (e.g., Bernier et al., 2019; Boldt et al., 2020; R. Feldman & Klein, 2003; Kochanska et al., 2000, 2008; Moore et al., 2009; Morelen et al., 2016), and across ages, from infancy to early childhood (Burgess et al., 2003), from early to middle childhood

(Norona & Baker, 2017), and from mid- to late adolescence (Van Lissa et al., 2019).

Child-to-parent effects are more nuanced, with possible developmental changes expected. At young ages, when children's emotion regulation is fledgling, parent-to-child effects may dominate. Parents serve as external resources for children, providing crucial support and assistance to regulate children's emotions (Baker, 2018; Eisenberg et al., 1998; Fox & Calkins, 2003; Kopp, 1989; Sroufe, 1996).

At older ages, child-to-parent effects may become apparent. Parents whose children are less well-regulated may continue to provide children with extra care to relieve their distress (Dadds & Roth, 2001) as part of adaptive bidirectional dynamics. But children whose emotion regulation capacities are well developed may lead parents to "pull back" and become less supportive. As children's regulatory capacities mature and their desire for autonomy rapidly grows, parents may recognize those developments and reduce their involvement in children's emotion regulation over time, offering help only when necessary (Eisenberg & Morris, 2003; Scaramella & Leve, 2004). For instance, children's better executive control at 36 months was related to mothers' lower levels of warmth but higher levels of scaffolding at 54 months, suggesting that parents likely decrease their involvement but encourage child autonomy when the child has better regulatory capacities (Klein et al., 2018).

Although such developmental changes in dynamics are expected, relevant longitudinal research is scarce, with only a handful of exceptions. R. Feldman (2015) examined longitudinal bidirectional relations between child emotion regulation and parent-child reciprocity at ages 1, 2, and 5 years and found that early parent-child reciprocity was related to children's better emotion regulation at the subsequent time point across all assessments. But children's emotion regulation predicted parent-child reciprocity only from age 2 to age 5. These findings suggest that parent-to-child effects may be stronger in infancy, but child-to-parent effects strengthen as children grow and develop autonomous regulatory strategies, consistent with our conceptualization.

To summarize, we studied the complex and developmentally changing dynamics of parent and child effects in the development of children's emotion regulation from infancy to early school age. Although we expected those dynamics to be bidirectional throughout development, we also expected differences between younger and older ages. We expected that parent-to-child effects—the impact of parental responsiveness on child future emotion regulation—would be present and beneficial throughout the studied age span. But we also expected that as children age, we would find more nuanced child-to-parent effects, with the parent tailoring their responsiveness to the child's regulatory capacities—providing more support to more emotionally dysregulated children but *less support* to those who were well regulated.

This study expands previous research in three important but underappreciated directions. First, we investigated these reciprocal relations for a long period from infancy to early school age, during which parent-child relationships are particularly critical in child development. Long-term longitudinal investigations of the transactional relations between parent and child effects are crucial because the current parent-child interactions can be influenced by their relationship histories (Cummings & Schermerhorn, 2003), and their dynamics can be altered over developmental time (Collins & Madsen, 2003; Maccoby, 1984b; Scaramella & Leve, 2004).

Second, we collected data from both mother– and father–child relations. Although scholars acknowledge that mothers and fathers may play different roles in children’s emotional development (Kiel & Kalomiris, 2015; Paquette, 2004; Power & Parke, 1982), our understanding of those differences is far from complete. Several researchers have demonstrated that both mothers’ and fathers’ positive parenting behaviors were related to better emotion regulation or relevant outcomes such as effortful control (Davidov & Grusec, 2006; Neppl et al., 2020). However, others have shown different patterns between mothers and fathers (Hurrell et al., 2015; Van Lissa et al., 2019), which may represent distinct contributions of mothering and fathering. Whereas maternal parenting focuses more on caregiving and comforting, father–child parenting often focuses on issues related to exploration. Those interactions can be more active, stimulating, and emotionally arousing, providing children with more opportunities to practice emotion regulation (Paquette, 2004; Parke, 1994; Volling et al., 2019). Such paternal “activation parenting” based on play-based interactions has been suggested to be particularly pertinent in promoting children’s regulatory capacities (J. S. Feldman & Shaw, 2021). Given the scarcity of relevant research and the unsettled state of our knowledge, more studies including both mothers and fathers are imperative to explore similarities or differences in the bidirectional associations between parenting and child emotion regulation.

And third—last, but certainly not least—we aimed for a more appropriate analytical approach to study longitudinal reciprocal associations between parental responsiveness and children’s emotion regulation within dyads. Traditionally, a cross-lagged panel model (CLPM) has been the most popular and common approach in longitudinal research. However, recently, the use of CLPM, especially in developmental research, has been criticized because CLPM conflates between-dyad and within-dyad effects (Berry & Willoughby, 2017; Hamaker et al., 2015). As such, CLPM cannot provide information regarding temporal variations in each dyad; rather, the paths in CLPM represent the stability of a person’s relative rank order in each variable (i.e., autoregressive paths) and the relations between the rank orders of one variable and others (i.e., cross-lagged paths).

This statistical issue elucidates and dovetails with an important conceptual issue, unique to developmental research on relationships, and one that, to our knowledge, has been rarely appreciated. When we are concerned with the dynamics within the parent–child relationship and specifically with processes that involve the parent and child adjusting their behavior based on their shared history, the CLPM approach misses an important point.

As mentioned earlier, the parent’s responsive care promotes the child’s emotion regulation because the child develops expectations of the parent’s comfort and support based on *their own experiences* with the parent (Dykas & Cassidy, 2011; Luo et al., 2024; Sherman et al., 2015). To the child, the parent’s rank order in responsiveness in the population is not relevant; what is relevant is the child’s unique, “lived,” or experienced history of the parent’s responsiveness to the child’s cues.

Likewise, the child’s emotion regulatory capacities influence the parent’s future response because the parent appraises the child’s psychological and developmental needs based on the *parent’s own experiences with the child*. To the parent, the child’s rank order in emotion regulation in the population is not relevant; the parent calibrates their responsiveness—up or down—based on their unique history with the child.

To address those unique qualities of parent–child mutual dynamics, we employed an autoregressive latent trajectory (ALT) model (Bollen & Curran, 2004, 2006). This model is much more appropriate and better suited to study how children and parents influence each other over time based on their past experiences because it separates within-dyad from between-dyad effects. In ALT, latent factors (e.g., intercept and slope) capture overall developmental trajectories of parental responsiveness and child emotion regulation from infancy to early childhood (between-dyad and traitlike components), and autoregressive and cross-lagged paths indicate assessment-to-assessment variations in each variable and their mutual relations, respectively (within-dyad and statelike components). A general developmental ALT model of adaptive bidirectional socialization dynamics between parent and child constructs is shown in Figure 1.

In sum, we aimed to examine longitudinal bidirectional associations between parental responsiveness and child emotion regulation from age 7 months to 6.5 years, using a statistical model separating within-dyad and between-dyad effects in both mother–child and father–child dyads. Our measures focused on children’s (dys-)regulation of negative emotion. Dysregulated negative emotion is broadly implicated as a key transdiagnostic factor contributing to the evolving maladaptive developmental trajectories (Beauchaine & Cicchetti, 2019). It is also easier to observe early in development than emotion regulation (Beauchaine et al., 2020). As well, the relation between parental responsiveness and child arousal was stronger following child negative than positive affect (Wass et al., 2019), dovetailing with attachment researchers’ emphasis on the utmost importance of responsiveness to child distress.

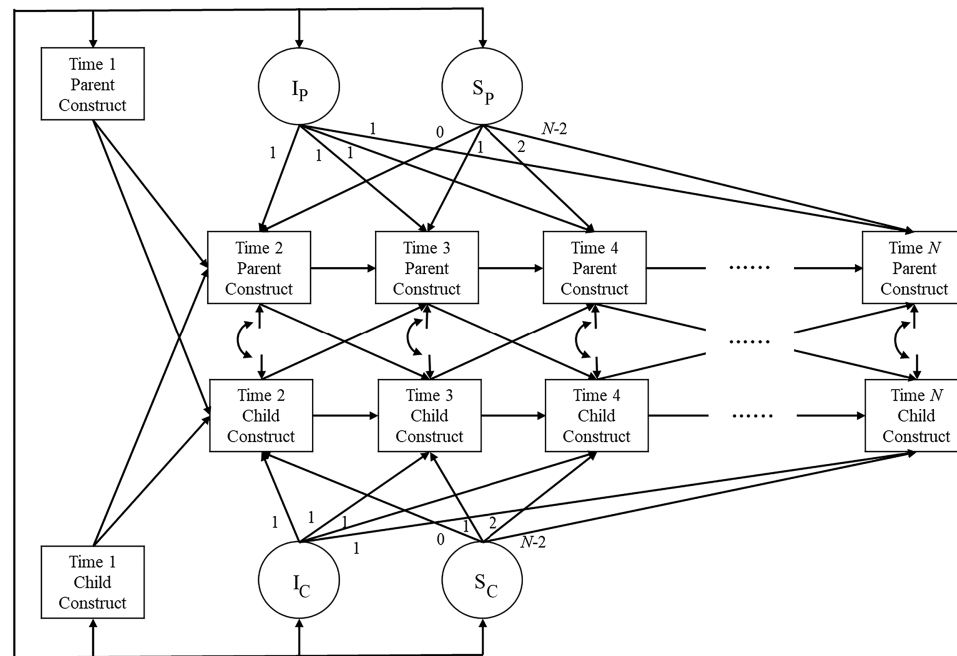
Method

Participants

The present study included typically developing 7-month-old infants, born mostly in 2001, and their biological parents (mothers and fathers). Families were recruited from a U.S. Midwestern area through flyers distributed in community venues and posts on social media targeting parent groups. The families were mostly White, but in 20% of the families, one or both parents were not White. Demographic details are in Supplemental Table S1.

Data were collected during 2- to 4-hr scripted sessions at home (at 7 months) and in naturalistic laboratory settings, including a Living Room and a Play Room (at the other time points), conducted by female experimenters. The sessions were parallel for both parents in terms of the scripts, and their order was randomized except that the mother–child session was performed first at 7 months. Data were coded from recorded videos by multiple coders. Coders used 15%–20% of cases for reliability, followed by regular realignments. Kappas (κ s), weighted κ s, and α s were used as appropriate (note that best practices have evolved over the last 2.5 decades). In this article, we report data from seven time points: 7 months, $N = 102$, 51 girls and 51 boys; 15 months, $N = 101$, 51 girls and 50 boys; age 2, $N = 100$, 50 girls and 50 boys; age 3, $N = 100$, 50 girls and 50 boys; age 4, $N = 99$, 49 girls and 50 boys; age 5.5, $N = 92$, 45 girls and 47 boys; and age 6.5, $N = 90$, 43 girls and 47 boys. Parents’ responsiveness and children’s emotion dysregulation were assessed at all times. Parents completed informed consent, and the institutional review board at the University of Iowa approved the study

Figure 1
Multivariate Autoregressive Latent Trajectory Model of Adaptive Bidirectional Socialization Dynamics Between Parent and Child Constructs



Note. Examples of parent constructs may include parental responsiveness, sensitivity, control, and so forth. Examples of child constructs may include child emotion regulation, compliance, antisocial conduct, and so forth. I = intercept; S = slope; P = parent construct; C = child construct.

(Developmental Pathways to Antisocial Behavior: A Translational Research Program, 200107049).

Measures

The Parent's Responsiveness to the Child

Parental responsiveness to the child was observed in carefully scripted naturalistic interactive contexts, such as introduction to the room, parent busy, snack, play, cleanup, and opening a gift, parallel for mother-child and father-child dyads, and standard for all dyads. The duration of each context varied from 2 to 15 min, and the total duration of the observations across all assessments ranged from 27 min (at age 3 years) to 65 min (at age 4 years). For each context, a global rating from 1 (*highly unresponsive*) to 7 (*highly responsive*) was given based on the parent's sensitivity-insensitivity, cooperation-interference, and acceptance-rejection (Ainsworth et al., 1971). Reliability, weighted κ s, ranged from .60 to .92. Cronbach's α s ranged from .90 to .98. The overall responsiveness score for the parent was averaged across all contexts at each time point. A higher score indicated more responsiveness. At all seven time points, mothers were more responsive to the child than fathers.

The Child's Emotion Dysregulation Expressed to the Parent

Child emotion dysregulation, expressed to the parent, was coded in the same contexts as parental responsiveness (with very minor

adaptations at 7 months). For each 30-s segment, coders coded the child's presentation of negative emotions as *not present*, *neutral negative* (neutral expression with a negative tint such as fatigue, impatience, and worry), *discrete* (full-blown expression of negative emotion such as distress, cry, and anger), or *intense* (strong or lasting more than 15 s) based on facial expressions, body language, and tone of voice, following extensive past research (An & Kochanska, 2023; Barry & Kochanska, 2010). Reliability, weighted κ s, ranged from .65 to .88.

We tallied the instances of each code, weighted each score by multiplying *neutral* by 1, *discrete* by 2, and *intense* by 3, and calculated a mean score by dividing the sum by the number of total segments. Higher scores indicated more dysregulated negative emotions. Children's emotion dysregulation scores with mothers and with fathers did not differ.

Covariates

We included children's gender and proneness to anger at 7 months as covariates. Proneness to anger was measured in the Car Seat episode from the Laboratory Temperament Assessment Battery (Goldsmith & Rothbart, 1996). The child was buckled in a car seat for 60 s, and their peak intensity of body anger was rated as 0 = *none*, 1 = *mild*, 2 = *medium*, 3 = *moderate*, 4 = *strong*; facial anger as 0 = *none*, 1 = *mild*, 2 = *moderate*, 3 = *strong*; and vocal anger as 0 = *none*, 1 = *mild*, 2 = *moderate*, 3 = *strong*. Reliability, κ s, ranged from .76 to .95. The overall intensity score was averaged across body, facial, and vocal anger.

Data Analytic Plan

We used an ALT model, which was developed to account for the limitations of the two most widely used approaches for longitudinal data (Bollen & Curran, 2004, 2006)—CLPM and a latent growth curve model. CLPM is useful for exploring how the prior level of one variable determines the later level of another variable while controlling for the stability of each variable. However, CLPM may bias cross-lagged effects because it does not account for individual traitlike stability or changes (Berry & Willoughby, 2017; Hamaker et al., 2015). The latent growth curve model does allow the investigation of separate trajectories of each individual over time, but not the relations among variables. ALT also differs from the random intercept CLPM (RI-CLPM; Hamaker et al., 2015), another widely used approach for modeling longitudinal data. While both ALT and RI-CLPM separate within-unit and between-unit variance, their primary goals differ. RI-CLPM focuses on capturing within-unit fluctuations around each individual's stable traitlike level by including random intercepts, but it does not account for developmental change over time. In contrast, ALT simultaneously models within-unit processes and trajectories of change (e.g., growth or decline), while also addressing unobserved heterogeneity. In sum, ALT, which integrates key features of CLPM and latent growth curve model and extends RI-CLPM, enables the study of *both* the developmental trajectories and the reciprocal associations.

In ALT, intercept and slope factors capture traitlike stability, and therefore, autoregressive paths and cross-lagged paths are based on statelike residuals (Bollen & Curran, 2004, 2006). Autoregressive paths do not represent the interindividual rank-order stability as in CLPM. Rather, they reflect within-person carry-over effects, or within-person predictions over time (i.e., the ability of statelike deviations of prior levels to predict later levels). For instance, a positive autoregressive coefficient of parental responsiveness implies

that if a parent scores higher than their own expected score in responsiveness to the child based on their own trajectory (note: not higher than the group mean) at the prior time point, they are likely to score higher than their own expected score at the subsequent time point. Likewise, cross-lagged paths represent within-dyad predictions over time. A negative cross-lagged coefficient from parental responsiveness to child emotion dysregulation implies that if a parent scores higher than expected in responsiveness to the child, the child is likely to score lower in emotion dysregulation expressed to the parent than the child's expected score at the subsequent time point.

Transparency and Openness

We report our recruitment procedures, measures, and analysis methods used. The analysis syntax can be found on the Open Science Framework (Kim, 2025; <https://osf.io/j938r>), and the coding systems are available upon request. But the data supporting the current findings are not publicly available. The consents our participants signed precluded any sharing of individual data, even if deidentified. This study's design and its analysis were not preregistered.

Results

Descriptive Statistics and Correlations

Descriptive statistics and correlations computed in SPSS 29 (IBM Corp., 2023) are presented in Table 1. In general, parental responsiveness and child emotion dysregulation were negatively and moderately correlated with each other. Concurrent relations were significant when children were younger (mothers: 7 months–2 years, fathers: 7 months–4 years), but not when they were older. For

Table 1

Descriptive Statistics of and Correlations Between Parental Responsiveness to the Child and the Children's Emotion Dysregulation

Construct	7 months	15 months	2 years	3 years	4 years	5.5 years	6.5 years	<i>M (SD)</i>
Child emotion dysregulation expressed to the mother								
Maternal responsiveness to the child								
7 months	–.28**	–.13	–.21*	–.23*	–.04	.004	–.06	4.82 (0.73)
15 months	–.07	–.23*	–.21*	–.32**	.08	.02	.09	4.95 (0.78)
2 years	–.13	.03	–.22*	–.24*	.07	.02	.14	4.87 (0.95)
3 years	–.14	–.02	–.20*	–.13	.11	.03	–.01	4.80 (0.66)
4 years	–.16	.05	–.19	–.10	.06	–.03	.12	5.07 (0.55)
5.5 years	–.10	.02	–.26*	–.09	.08	.03	.003	5.04 (0.77)
6.5 years	–.10	.13	–.09	–.02	.13	.03	.11	4.75 (0.97)
<i>M (SD)</i>	0.38 (0.34)	0.21 (0.24)	0.13 (0.23)	0.08 (0.17)	0.06 (0.08)	0.03 (0.11)	0.07 (0.19)	
Child emotion dysregulation expressed to the father								
Paternal responsiveness to the child								
7 months	–.29**	–.21*	–.17	–.17	–.15	–.01	–.02	4.44 (0.93)
15 months	–.18	–.23*	–.03	–.06	–.29**	–.07	.05	4.51 (0.93)
2 years	–.12	–.04	–.30**	–.20*	–.26**	–.11	–.13	4.44 (0.98)
3 years	–.13	–.07	–.06	–.27**	–.21*	–.03	–.15	4.47 (0.95)
4 years	–.13	–.25*	–.19	–.08	–.32**	–.06	.002	4.79 (0.69)
5.5 years	–.03	–.08	–.15	–.26*	–.23*	–.23	–.32**	4.64 (0.81)
6.5 years	–.17	–.11	–.13	–.20	–.16	.02	–.07	4.29 (0.93)
<i>M (SD)</i>	0.40 (0.41)	0.19 (0.23)	0.14 (0.24)	0.08 (0.15)	0.07 (0.13)	0.03 (0.13)	0.02 (0.05)	

* $p < .05$. ** $p < .01$.

longitudinal relations, those from early parental responsiveness to later child emotion dysregulation were more robust than vice versa.

Relations Between Parental Responsiveness and Child Emotion Dysregulation

The models were estimated in Mplus 7 (Muthén & Muthén, 1998–2012) with maximum likelihood with robust standard errors estimator. As in CLPM, in which the initial measurement is set as a predetermined exogenous variable, we treated the first measurement of each variable as predetermined and covaried them with the latent curve factors (i.e., intercepts and slopes). It enabled us to estimate autoregressive and cross-lagged paths without bias from omitted prior measures of variables because their effects are considered to be absorbed into the initial exogenous measures (Bollen & Curran, 2004). Then, the latent curve factors were created using measurements from the second and later time points.

We followed the steps in Bollen and Curran (2004) to build the most parsimonious ALT model that best fits the data. First, we tested a univariate ALT model for each variable by comparing the model fits of the growth model with both intercept and slope included, the growth model without slope variance, and the intercept-only model. Then, we estimated the full bivariate model with intercepts and slopes included for both variables without any constraints. As in univariate model testing, we first compared the full model with the one without slope variance and then the intercept-only model. Once proper latent curve factors were obtained, we gradually applied equality constraints to within-time covariance, to autoregressive paths of each variable, and to cross-lagged paths. Nested models were compared using the Satorra–Bentler scaled chi-square difference test (Satorra & Bentler, 2001). If the model fit significantly differed between the two models, the more complex model (with more estimated parameters) was retained. If not, the more parsimonious model (with more constraints) was retained. Once the most parsimonious model was retained, we included covariates to create the final model.

The final model fit was evaluated using the comparative fit index (CFI) and the root-mean-square error of approximation (RMSEA) and its 90% confidence interval. Model fit is considered good when CFI is larger than or equal to .95 and RMSEA is less than or equal to .05 and mediocre or acceptable when CFI is larger than or equal to .85–.90 and RMSEA is less than or equal to .08 (Hu & Bentler, 1999; Little, 2013). The final model fit was acceptable in both mother–child (CFI = .83, RMSEA = .08, 90% confidence interval [.06, .10]) and father–child dyads (CFI = .90, RMSEA = .07, 90% confidence interval [.04, .09]).

Mother–Child Dyads

The results of model-building processes are in Supplemental Table S2. The slope factor of maternal responsiveness and the slope variance of child emotion dysregulation were removed without significantly changing the overall model fit. Within-time covariances, autoregressive paths of child emotion dysregulation, and cross-lagged paths from child emotion dysregulation to maternal responsiveness were constrained equal over time.

The final multivariate model is presented in Figure 2. We found no significant effects of child gender or proneness to anger. The average initial level of maternal responsiveness at 7 months was

high ($M = 4.83$ on a 1–7 scale), with significant between-person variability ($S^2 = 0.53$, $p < .001$). The average maternal responsiveness at 15 months, after accounting for the initial level of maternal responsiveness at 7 months, was 4.64 (i.e., intercept factor mean), and it significantly differed among mothers (i.e., significant intercept factor variance; $S^2 = 0.23$, $p < .001$). On average, maternal responsiveness did not change over time, as we were able to retain the model without a slope factor. But there were no significant within-person predictions over time in maternal responsiveness, as autoregressive paths were not significant.

The average initial level of child emotion dysregulation at 7 months was low ($M = 0.39$ on a 0–3 scale) with significant between-person variability ($S^2 = 0.12$, $p < .001$). Intercept factor mean indicated that the average of emotion dysregulation at 15 months, with the initial level of emotion dysregulation controlled, was 0.30, and it did not significantly differ among children. On average, emotion dysregulation declined by 0.05 per year (i.e., mean of slope factor), although only marginally significant ($p = .08$). This decline applied to all children, without significant between-person variability as the slope variance was fixed to 0. We found no within-person prediction over time in emotion dysregulation (i.e., no significant autoregressive paths).

We found few concurrent relations between maternal responsiveness and child emotion dysregulation. The initial levels of maternal responsiveness and emotion dysregulation at 7 months were significantly related. However, there was no significant relation between traitlike maternal responsiveness and emotion dysregulation at 15 months, after accounting for their initial levels at 7 months (i.e., nonsignificant relations between intercepts). In addition, we found no within-time covariances between disturbances of maternal responsiveness and emotion dysregulation from 15 months to 6.5 years, although they were marginally significant and negative, $B = -0.01$, $SE = 0.003$, $p = .05$.

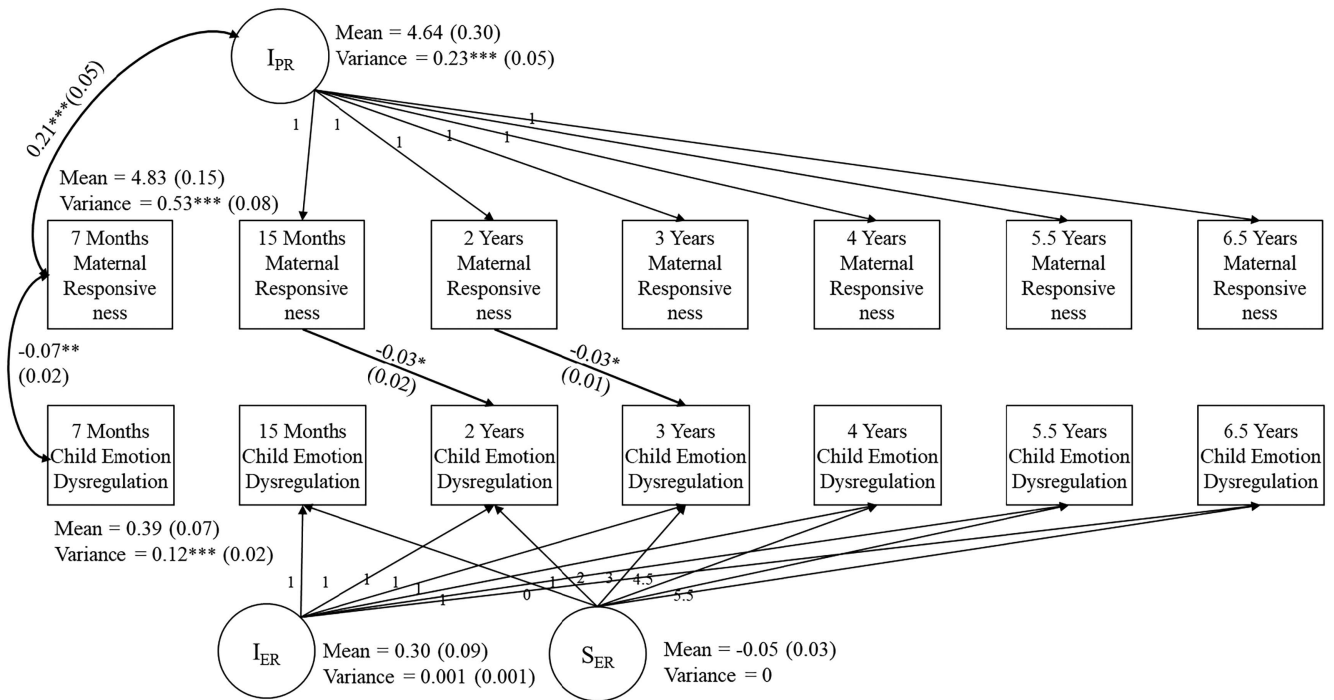
For longitudinal relations, maternal responsiveness at 15 months and 2 years negatively predicted emotion dysregulation at 2 and 3 years, respectively. Thus, if mothers were more responsive than expected (based on their trajectory of responsiveness), their children demonstrated more regulated negative emotions at the subsequent time point, particularly in toddlerhood. However, we found no significant relations from maternal responsiveness at 3 years and later to child emotion dysregulation at the corresponding subsequent time point.

On the other hand, the opposite direction (i.e., prior child emotion dysregulation predicting later maternal responsiveness) was not significant at any time points. This suggests that in the parent–child dynamics we studied, mother-to-child effects may be stronger than child-to-mother effects, at least in toddlerhood.

Father–Child Dyads

The results of model-building processes are in Supplemental Table S3. As in the mother–child model, the slope factor of paternal responsiveness and the slope variance of child emotion dysregulation were removed without significantly changing the overall model fit. Autoregressive paths of emotion dysregulation were constrained to be equal across time points.

The final results are in Figure 3. We found no significant gender effects, but proneness to anger at 7 months was significantly

Figure 2*Autoregressive Latent Trajectory Model of Relations Between Maternal Responsiveness and Child Emotion Dysregulation*

Note. Unstandardized coefficients and standard errors (in parentheses) are presented for significant paths only. Child gender (0 = girls, 1 = boys) and anger proneness at 7 months were included in the analysis but not depicted in the figure for clarity. I = intercept; S = slope; PR = parental responsiveness; ER = child emotion dysregulation.

* $p < .05$. ** $p < .01$. *** $p < .001$.

related to the initial level, intercept, and slope of child emotion dysregulation. Children with higher temperamental anger exhibited more poorly regulated negative emotions, and their change in emotion dysregulation was more stable. The average initial level of paternal responsiveness was high ($M = 4.67$ on a 1–7 scale), with significant between-person variability ($S^2 = 0.84$, $p < .001$). Intercept factor mean indicated that the average paternal responsiveness at 15 months, after accounting for the initial level of paternal responsiveness at 7 months, was 3.58, and it significantly differed among fathers ($S^2 = 0.17$, $p < .01$). On average, paternal responsiveness was stable over time as we retained the model without slope factor. We found significant positive autoregressive paths of paternal responsiveness at all time points. Thus, fathers who were higher than expected in responsiveness (based on their trajectory of responsiveness) showed higher-than-expected responsiveness at the subsequent time point.

The average initial level of child emotion dysregulation expressed to the father was low ($M = 0.14$ on a 0–3 scale), with significant between-person variability ($S^2 = 0.15$, $p < .01$). Intercept factor indicated that the average of child emotion dysregulation at 15 months, after accounting for the initial level of child emotion dysregulation, was 0.25, and it was significantly different among children ($S^2 = 0.003$, $p < .05$). On average, emotion dysregulation declined by 0.03 per year, and this decline applied to all children, without significant between-person variability as the slope variance was fixed to 0. Interestingly, we found significant negative

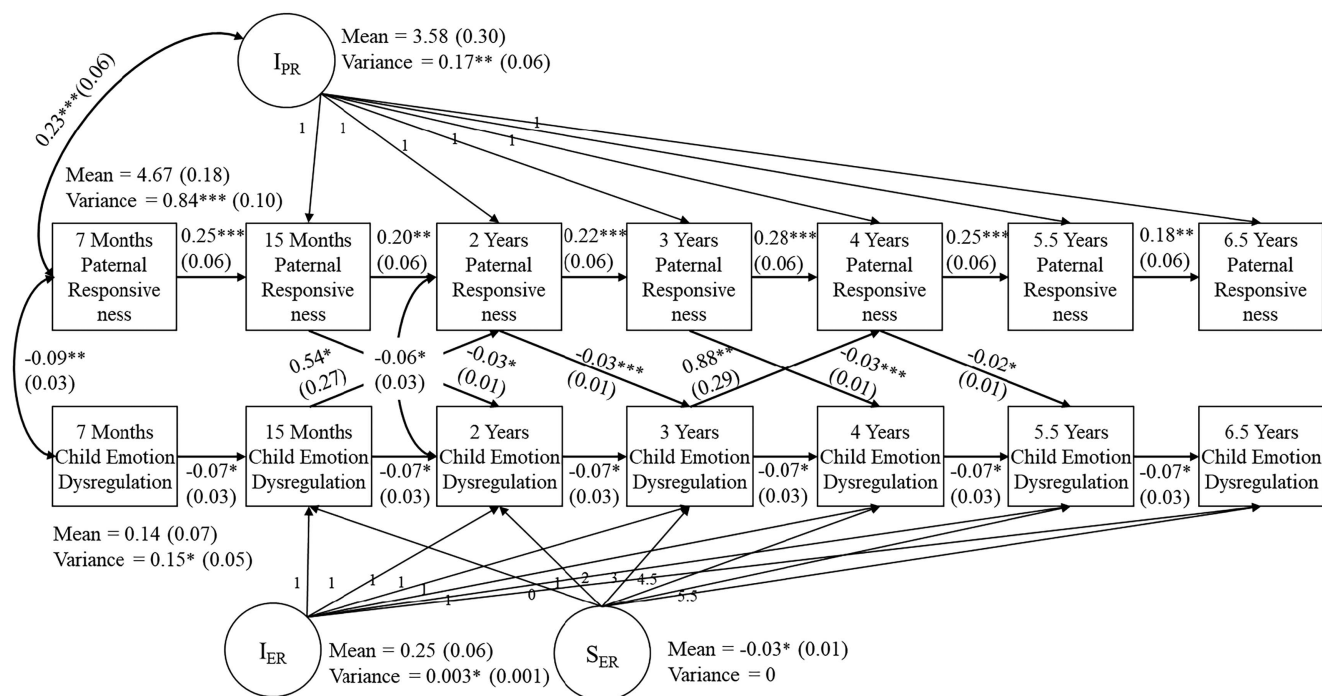
within-person prediction over time (i.e., negative autoregressive paths). Thus, children with higher-than-expected emotion dysregulation at the prior time point had lower-than-expected emotion dysregulation at the subsequent time point.

Within-time covariances between paternal responsiveness and emotion dysregulation were not constrained equal over time, and we found that higher paternal responsiveness was concurrently related to lower emotion dysregulation at 7 months and 2 years. There were no other significant concurrent relations.

We found several significant longitudinal relations. Paternal responsiveness at 15 months, 2, 3, and 4 years negatively predicted child emotion dysregulation at 2, 3, 4, and 5.5 years, respectively. Thus, if fathers were more responsive than expected, their children demonstrated fewer difficulties in emotion regulation at the subsequent time point. We found no significant relations from paternal responsiveness at 7 months and 5.5 years to child emotion dysregulation at the subsequent time point.

In addition, if children demonstrated higher-than-expected emotion dysregulation at 15 months and 3 years, again based on the trajectory of child's emotion dysregulation, their fathers were more responsive than expected at 2 and 4 years, respectively. We found no significant child-to-father effects at the other time points.

Thus, compared to mother–child dyads, in which we found only mother-to-child effects in toddlerhood, father–child dyads showed more co-regulatory dynamics, especially during toddlerhood and

Figure 3*Autoregressive Latent Trajectory Model of Relations Between Paternal Responsiveness and Child Emotion Dysregulation*

Note. Unstandardized coefficients and standard errors (in parentheses) are presented for significant paths only. Child gender (0 = girls, 1 = boys) and anger proneness at 7 months were included in the analysis but not depicted in the figure for clarity. I = intercept; S = slope; PR = parental responsiveness; ER = child emotion dysregulation.

* $p < .05$. ** $p < .01$. *** $p < .001$.

early childhood. However, father-to-child effects seemed stronger than child-to-father effects.

Discussion

Parent-child relationships provide foundational contexts for the development of emotion regulation in early life. Although the reciprocal link between parenting and child emotion regulation has been well acknowledged, the extent to which this association remains unchanged or altered across developmental periods is underappreciated and poorly understood. In particular, the role of fathers in such developmental dynamics has received relatively little attention, although its importance was proposed over 4 decades ago (Power & Parke, 1982), and research continues to imply that fathers help children regulate emotional arousal by encouraging risk-taking play during which children can practice regulatory skills (Grossmann et al., 2002; Paquette, 2004; Puglisi et al., 2024; Volling et al., 2019).

To address these gaps, the present study examined the bidirectional associations between parental responsiveness and child emotion dysregulation from infancy to early school age, in both mother- and father-child dyads. We hypothesized that parent-to-child effects would be stronger than child-to-parent effects at younger ages, when children rely on external support for regulation and learn adaptive skills from their parents. As children's regulatory capacities mature, we expected child-to-parent effects to strengthen, and the dynamics to become increasingly bidirectional. Our findings

provided partial support for these hypotheses. Moreover, some of the patterns in mother- and father-child dyads were similar, but some appeared quite distinct.

First, we found evidence of parent-to-child effects in toddlerhood in both dyads and in early childhood in father-child dyads only. As hypothesized, children of mothers and fathers who engaged in higher-than-expected responsiveness displayed lower-than-expected emotion dysregulation at the subsequent time point, whereas children of parents who engaged in lower-than-expected responsiveness displayed higher-than-expected emotion dysregulation. These findings align with previous research linking responsive and supportive parenting to more adaptive regulatory skills and unresponsive or inconsistent parenting to greater emotion dysregulation (e.g., Beliveau et al., 2023; Goagoses et al., 2023; Paulussen-Hoogeboom et al., 2007; Wass et al., 2024). Consistent and responsive parenting fosters children's mental representations of the parent as a safe haven and secure base and promotes effective, secure-base emotion regulation strategies when distressed (Cassidy et al., 2025). On the other hand, children who receive less-than-expected responsiveness may exaggerate their emotions to elicit the parent's attention. Wass et al. (2025) found that children exposed to unresponsive parenting were more likely to vocalize their negative emotions even when their physiological arousal level was low ("shouting to be heard," p. 2415).

The attenuated maternal effects may reflect nuances in how maternal responsiveness is perceived and measured. Given that mothers are more likely to play a child-rearing role than fathers

(Paquette, 2004; Volling et al., 2019) and, in our sample, mothers spent more time with children than fathers (i.e., 8.2 hr/day for mothers and 4.3 hr/day for fathers), children may perceive maternal responsiveness as a “default” and be less affected by its fluctuation. It is also likely that significance of maternal responsiveness would be better captured in contexts highly distressing to the child; those were limited in our sessions, and most of our laboratory contexts were relatively positive, which may have aligned better with the oft-described qualities of father–child interactions (Grossmann et al., 2002; Paquette, 2004; Volling et al., 2019).

Second, we found child-to-parent effects from 15 months to 2 years and from 3 to 4 years, supporting the bidirectionality, but—surprisingly—only in father–child dyads. Consistent with our hypothesis, children with greater emotion dysregulation received more responsiveness from their fathers than expected at the subsequent time point, whereas those with better early emotion regulation received less responsiveness than expected. Fathers of well-regulated children may decrease their involvement to encourage autonomy (Eisenberg & Morris, 2003; Scaramella & Leve, 2004), whereas fathers of less well-regulated children may increase their responsiveness to support their child’s emotional needs (Dadds & Roth, 2001). This aligns with the previous research showing that toddlers with dysregulated fear were more likely to receive more protective parenting, although that study only included mothers (Kiel & Buss, 2014).

This pattern of findings is compelling and innovative, because it suggests that fathers may be more attuned to their children’s regulatory needs than mothers. Although more research is needed to explain this intriguing pattern, evidence consistent with our findings has been reported. In RI-CLPM, examining the longitudinal relations between emotion-related parenting and child negative affect in mother–child relationships, no significant within-dyad cross-lagged effects were found (Price et al., 2025). In another study on the relations between child characteristics and maternal and paternal parenting, McBride et al. (2002) found that fathers calibrated their involvement to the child’s temperamental characteristics more than mothers did. The authors proposed that this discrepancy may stem from different societal expectations surrounding parenting roles. Although the importance of fathering has gained recognition in recent years, mothers are still widely viewed as primary caregivers, and they may feel constrained by societal norms and expected to “do their best” regardless of the individual child’s characteristics. But fathers, who presumably face fewer societal expectations, may feel freer to be flexible in their parenting and tailor their behavior to their child’s characteristics. In our sample, mothers appeared to play primary caregiver roles in general, because the time mothers spent with their children was almost double that of fathers on average. Interestingly, the time with children was not related to the level of parental responsiveness. Thus, different patterns of findings between mothers and fathers in our results are likely due to different societal expectations imposed on mothers and fathers, not due to the time with children per se. Note, however, that the few studies of father–child bidirectional dynamics have produced mixed results (Taraban & Shaw, 2018).

Third, we noted an absence of significant longitudinal relations between parental responsiveness and child emotion dysregulation in infancy and early school ages. The null findings in infancy were surprising and inconsistent with our expectations. However, they partly align with R. Feldman’s (2015) findings that the associations

between parent–child reciprocity and child emotion regulation were weak from ages 1 to 2 but became stronger from ages 2 to 5. R. Feldman suggested that relations between parenting and child emotion dysregulation in infancy are still in flux and may be affected by other contextual or dispositional factors. For example, such relations can be present only for children with insecure attachment but not for those with secure attachment (An & Kochanska, 2023). Further, the stability of emotion dysregulation may vary for secure and insecure children (Brock & Kochanska, 2019).

The null effects in early school ages are consistent with the findings in previous research showing that the relations between parental supportive socialization and child socioemotional outcomes such as emotion regulation, social competence, and behavioral problems were significant among children younger than age 4 but not those older than age 5 (Mirabile et al., 2018). Such findings likely reflect children’s growing competencies in emotion regulation. As children develop more sophisticated regulatory strategies, they may require less parental guidance, weakening the reciprocal links between parenting and emotion regulation. The negative slope of emotion dysregulation in both dyads supports this interpretation, indicating that children improved in regulating their negative emotions over time.

In addition, the null findings in early school age may be partly because children’s everyday ecologies and interpersonal relationships expand considerably at the entry of formal schooling. As such, new contextual factors such as peer relations and school climate may become more prominent influences on emotion regulation, while the effects of parenting diminish (Baker, 2018). Likewise, parents may become less affected by momentary fluctuations in their child’s emotional expressions. Rather, external stressors outside of family, such as stressful environments, work demands, and marital satisfaction, may exert greater effects on parental responsiveness (Collins & Madsen, 2003; Ponnet et al., 2013).

Finally, we failed to find significant within-time covariances between parental responsiveness and child emotion dysregulation after accounting for the initial levels at 7 months and overall trajectories, although they were marginally negative in mother–child dyads, and we found one significant relation at age 2 years in father–child dyads. This suggests that the link between parental responsiveness and child emotion dysregulation is primarily longitudinal rather than concurrent. As their experience with each other accumulates, parents and children may gradually reorient and adjust their interactions accordingly.

In addition, the absence of significant within-time covariances can highlight the importance of separating between-dyad and within-dyad processes. Between-dyad and within-dyad effects are conceptually different, and their magnitudes and directions can also differ (Curran & Bauer, 2011; Hamaker et al., 2015). Thus, significant within-time relations observed in previous research using a traditional CLPM, which conflates within- and between-dyad effects, may not necessarily replicate when these effects are disentangled.

Overall, our findings highlight fathers’ crucial role in children’s developing emotion regulation—perhaps more critical than mothers’. Research has emphasized the role of play-based interactions, particularly with fathers, in socioemotional development, as these interactions require children to manage emotions and respond to social cues dynamically to maintain activities (Leidy et al., 2013; Paquette, 2004). Supporting this idea, growing evidence has shown the importance of fathering in emotion regulation development.

Fathers' parenting had unique associations with children's regulatory abilities over and above maternal influences (R. Feldman & Klein, 2003; Karreman et al., 2008; McElwain et al., 2007), and parents' emotional withdrawal was related to children's less adaptive emotion regulation in father-child, but not in mother-child interactions (Gallegos et al., 2017). In addition, the patterns of bidirectional dynamics may depend on both parent and child gender, given that parent-child relationships can differ between mother-daughter, mother-son, father-daughter, and father-son dyads, although the results were mixed (Chang et al., 2003; R. Feldman, 2003; Lovas, 2005; Russell & Saebel, 1997). In our model, we were unable to compare the results between the four dyads due to a relatively small sample size, but we found no significant child gender effects in both mother-child and father-child dyads. Further research with both mothers and fathers ought to explore specific factors or mechanisms that can explain different patterns in this changing dynamic of reciprocal relations between mother-child and father-child relationships.

These findings have important practical implications for parenting interventions and family-based programs. According to a recent review article on three main parental emotion socialization interventions, such as tuning in to kids, parent-child interaction therapy-emotion development, and emotion enhanced triple P (England-Mason & Gonzalez, 2020), these programs effectively improved the quality of parenting. However, findings regarding their effectiveness in encouraging child emotion regulation were limited, and most programs predominantly involved mothers. As such, more intervention research should include both mothers and fathers and properly assess the associations between parenting and child emotion regulation. Encouraging fathers' active involvement in children's emotional development may considerably enhance co-regulation processes and promote adaptive regulatory skills. Further, our data suggest that a common view of mothers as superior in their "natural" ability to adapt and calibrate their parenting to the child's developmental changes may be unfounded, and their sensitivity should also be targeted, especially with highly dysregulated children. Interventions aimed at improving parental responsiveness should target parents' appreciation of developmental processes, ensuring that parents are equipped to support children's growing autonomy while remaining attuned to their emotional needs. Future research should further explore the specific mechanisms underlying these dynamics with diverse populations to inform targeted strategies for fostering healthy parent-child relationships and emotion regulation outcomes.

This study has several limitations. First, our sample consisted of low-risk families, in which parental responsiveness was high and stable and child emotion dysregulation was low and decreased on average. Patterns in socioeconomically or contextually disadvantaged populations may differ, where greater environmental stressors may disrupt responsive parenting and exacerbate emotion regulation difficulties (Paulussen-Hoogeboom et al., 2007; Price et al., 2025; Wass et al., 2025). Also, children with elevated symptoms of psychopathology may exhibit different regulatory patterns and parent-child dynamics (He et al., 2023). Future research with more diverse familial and contextual backgrounds and children with regulatory problems, such as excessive irritability, will be important.

Second, we assessed responsiveness and emotion dysregulation using relatively global behavioral codes, but it is important to

recognize that those constructs are multidimensional. Different aspects of parental responsiveness, such as cooperation, positivity, sensitivity, or emotional availability, may have distinct associations with child emotion regulation (Bernier et al., 2021). Likewise, different types of child negative emotions, such as sadness, anger, and fear, may have different relations with parenting (Cole, 2014; Cole et al., 2009; Kim & Kochanska, 2024; Morris et al., 2007). Future research should explore more specific and nuanced dimensions of parental responsiveness and child emotion to provide a more comprehensive picture of the associations between parental responsiveness and child emotion regulation.

And third, features of parenting other than responsiveness should be considered in their interplay with child emotion regulation. For example, parental control, both positive and negative, had stronger associations with child regulatory skills than parental responsiveness (Karreman et al., 2006), and different components of mother-child negative interactions, such as initiation, length, frequency, and the tendency to end the negative cycle, had different relations with child negative affect (Moed, 2022). Future research with various aspects of parenting, including responsiveness and control, and their multiple components, can provide a more comprehensive and more nuanced understanding of how parenting impacts and is impacted by child emotion regulation.

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Received April 29, 2025

Revision received July 29, 2025

Accepted August 7, 2025 ■